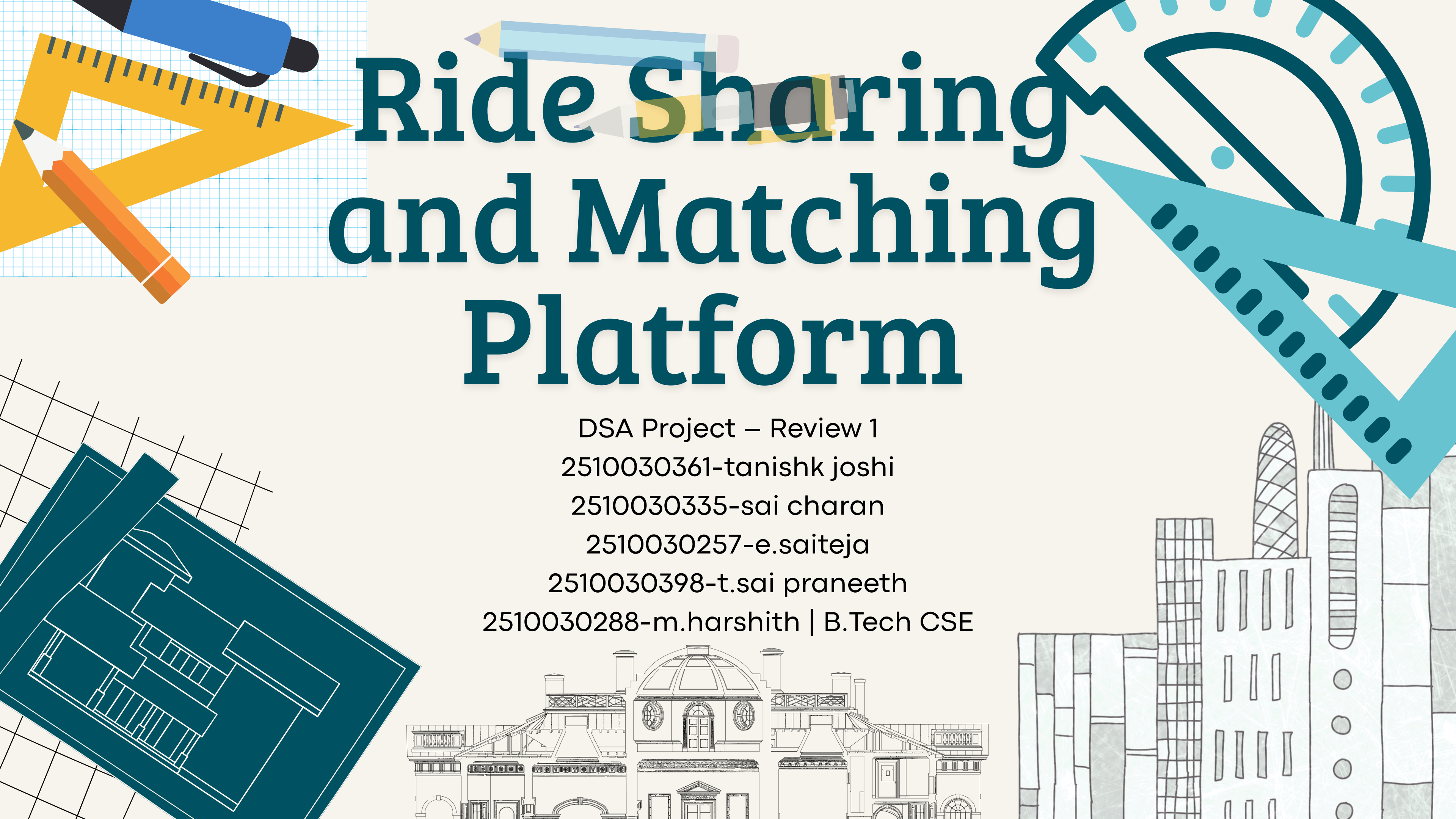




Ride Sharing and Matching Platform

DSA Project – Review 1

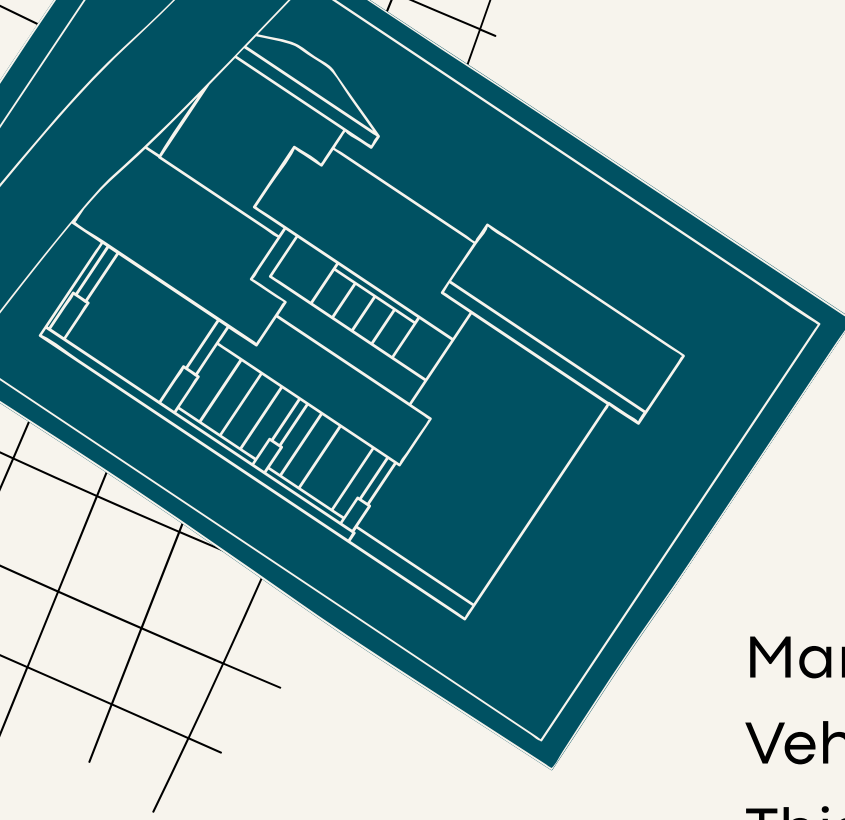
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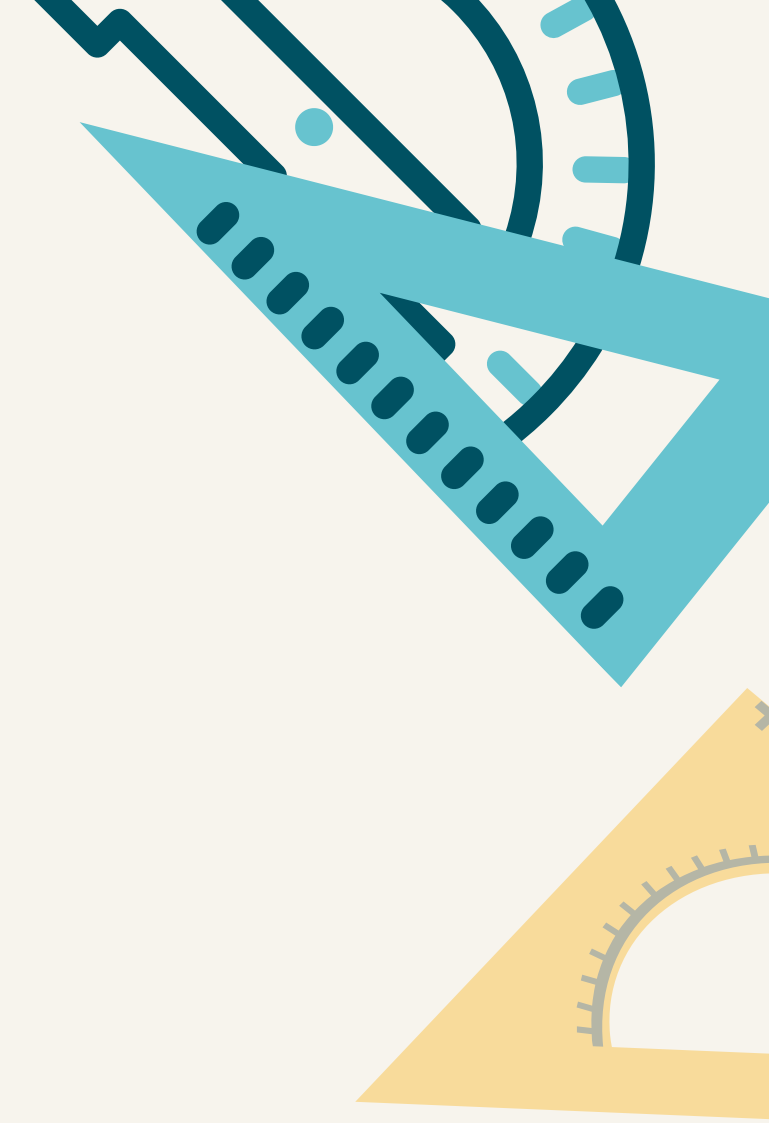
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2510030398-t.sai praneeth

2510030288-m.harshith | B.Tech CSE

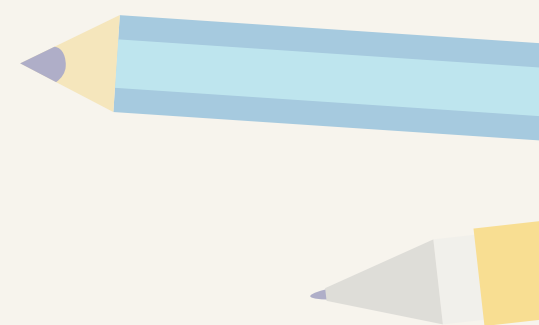


Problem



Many people travel alone daily with empty seats in their vehicles
Vehicles have empty seats that go unused on daily commutes
This leads to wasted resources and unnecessary travel costs

Fuel costs are high and keep increasing for individual travelers
Traffic congestion and pollution are growing problems in cities
👉 We need a smart system to match and share rides efficiently





Our Idea

The Platform

Create a ride-sharing platform where users can offer or request rides. The system intelligently matches people traveling on similar routes. Users simply enter their start and end locations to get started.

Key Benefits

Saves money by splitting fuel costs among riders. Reduces traffic congestion and pollution. Saves time with optimized route matching.

💡 Simple idea: "Find people going the same way"

Working

Step 1 & 2

User enters start and end location into the platform.
System checks all available rides in the database.

👉 Matching is based on route + time

Step 3 & 4

Matches users with similar route and timing.
Shows best ride options to the user for selection.

👉 Smart matching saves time and fuel cost

DSA Concepts

- Graph (for routes and locations)
- Searching (to find available rides)
- Sorting (best rides shown first)
- HashMap (to store user data efficiently)

👉 These DSA concepts help make the ride-sharing system fast and efficient. Graph structures model real-world routes, while Searching and Sorting algorithms quickly match riders. HashMap enables constant-time data access for user profiles and ride records.